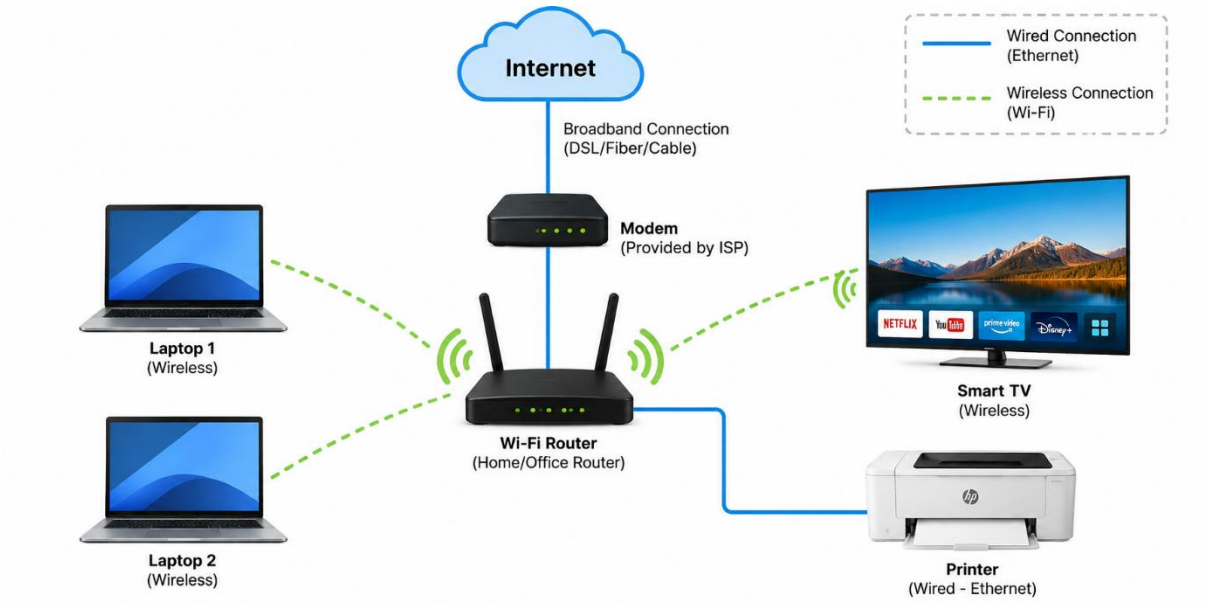
# Build a SOHO network

1. Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).




Here is the SOHO network diagram with all devices and connection types clearly labelled.

- Broadband Internet → Modem → Wi-Fi Router: Wired connection
- Wi-Fi Router → Laptop 1: Wireless (Wi-Fi)
- Wi-Fi Router → Laptop 2: Wireless (Wi-Fi)
- Wi-Fi Router → Smart TV: Wireless (Wi-Fi)
- Wi-Fi Router → Printer: Wired (Ethernet)

2. List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed.

| No. | Hardware Component | Suggested Brand/Model | Purpose |
|---|---|---|---|
| 1 | Broadband Modem/ONT | ISP-provided modem/ONT | Connects the network to the internet. |
| 2 | Wi-Fi Router | TP-Link Archer C6 | Provides Wi-Fi and connects devices to the network. |

| | | | |
|---|---|---|---|
| 3 | Network Switch (optional) | TP-Link TL-SG105 | Connects multiple wired devices. |
| 4 | Ethernet Cables | Cat6 RJ45 cables | Connects the router, switch, and computers. |
| 5 | Desktop/Laptop | Dell, HP, or Lenovo | Allows users to access the network and internet. |
| 6 | Wi-Fi Extender (optional) | TP-Link RE315 | Extends Wi-Fi coverage to distant areas. |
| 7 | Network Adapter | Built-in Ethernet/Wi-Fi adapter | Enables devices to connect to the network. |
| 8 | Power Supply/Adapter | Supplied with each device | Provides power to the networking equipment. |

3. Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

**Step-by-Step Wi-Fi Router Configuration for a SOHO Network**

1. Connect the router: Connect the router to the modem/ONT using an Ethernet cable and switch on both devices.

2. Connect to the router: Connect your computer to the router using an Ethernet cable or its default Wi-Fi network.

3. Open the admin page: Open a web browser and enter the router's management address, such as 192.168.0.1 or 192.168.1.1. Check the router label or manual for the correct address.

4. Log in: Enter the default administrator credentials provided with the router.

5. Change the admin password: Open the Administration or System settings and set a strong, unique password. Save the changes.

6. Set a unique SSID: Open Wireless Settings and enter a recognizable Wi-Fi name, such as Harsh_SOHO_Network.

7. Enable WPA2 security: Under Wireless Security, select WPA2-PSK (AES) or WPA2-Personal (AES) and set a strong Wi-Fi password.

8. Save the settings: Click Save or Apply. The router may restart, and your device may disconnect.

9. Reconnect to Wi-Fi: Select the new SSID and enter the new Wi-Fi password.

10. Test the connection: Open a website to confirm internet access. Connect another device to verify that the network works correctly.

11. Verify security: Log back into the router and confirm that WPA2 security is enabled and the default administrator password has been changed

4. Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

Use the network 192.168.1.0/24 with subnet mask 255.255.255.0. Assign static IP addresses to the printer and smart TV, and use DHCP to assign IP addresses automatically to the laptops.

| Device | MAC Address (Sample) | IP Address | Configuration |
|---|---|---|---|
| Wi-Fi Router | 00:1A:2B:3C:4D:01 | 192.168.1.1 | Static |
| Network Printer | 00:1A:2B:3C:4D:02 | 192.168.1.10 | Static |
| Smart TV | 00:1A:2B:3C:4D:03 | 192.168.1.11 | Static |
| Laptop 1 | 00:1A:2B:3C:4D:04 | 192.168.1.100 | DHCP |
| Laptop 2 | 00:1A:2B:3C:4D:05 | 192.168.1.101 | DHCP |

Configuration steps:

1. Log in to the router's admin page.
2. Set the router's LAN IP address to 192.168.1.1.
3. Enable DHCP and set the address pool to 192.168.1.100–192.168.1.200.
4. Configure the printer with IP 192.168.1.10 and the smart TV with IP 192.168.1.11, using subnet mask 255.255.255.0 and default gateway 192.168.1.1.
5. Set both laptops to obtain IP addresses automatically through DHCP.
6. Save the settings and test connectivity by pinging the router from a laptop.

5. Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

If a laptop cannot connect to the internet, I would check the following possible causes and solutions.

| No. | Possible Cause | Diagnosis and Solution |
|---|---|---|
| 1 | Wi-Fi is disabled or disconnected | Check Wi-Fi settings, enable Wi-Fi, and reconnect to the correct SSID using the correct password. |

| | | |
|---|---|---|
| 2 | Incorrect IP configuration | Run ipconfig in Command Prompt. Set the laptop to obtain an IP address automatically through DHCP. |
| 3 | Router or modem issue | Restart the router and modem. Check the indicator lights and verify that the internet connection is working. |
| 4 | DNS problem | Run ipconfig /flushdns. Test using ping 8.8.8.8 and nslookup google.com to identify DNS issues. |
| 5 | Weak Wi-Fi signal or interference | Move the laptop closer to the router or connect using an Ethernet cable to test connectivity. |